1st Place: Aspect 4 – Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **Analysis**: This is the first and most fundamental step for students approaching the question. They must read the tabular data (average daily production, average percentage of faulty players) and three statements, and understand the meaning of each statement within the specific context (Electrix Company's production).
- **Manifestation in the Question**: Students need to decode phrases such as "one third," "exactly 5 out of 100," and "probability...is 0.03," and connect them to the specific numbers in the table (2000, 6000, 5%, 3%). This allows them to mentally construct a mathematical model concerning production distribution and fault rates.

2nd Place: Aspect 1 – Select a mathematical description or a representation that describes a problem

- **Analysis**: After understanding the context, students need to select appropriate mathematical tools or concepts to evaluate each statement.
- **Manifestation in the Question**:
  - For Statement 1 (whether video players constitute one third of daily production), the mathematical description "proportion/fraction" must be selected: proportion of video players = 2000 / (2000+6000) = 1/4, not 1/3.
  - For Statement 2 (whether exactly 5 out of every 100 video players are faulty), the description "meaning of percentage" must be selected: 5% is an average over time, not a guarantee for every batch of 100.
  - For Statement 3 (whether the probability that a randomly chosen audio player needs repair is 0.03), the description "statistical definition of probability" must be selected: a 3% fault rate can be directly interpreted as the probability of a single randomly selected item being faulty.

3rd Place: Aspect 6 – Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **Analysis**: Students need to abstract the key mathematical variables from the production context and understand the relationships between them.
- **Manifestation in the Question**: The key variables include: daily production of video players (`V=2000`), daily production of audio players (`A=6000`), video fault rate (`p_v=5%`), and audio fault rate (`p_a=3%`). Correct judgment of the statements relies on accurately identifying and applying these variables and their meanings.

4th Place: Aspect 5 – Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

- **Analysis**: The question requires students to see the underlying mathematical relationships and patterns in the data.
- **Manifestation in the Question**:
  - Statement 1 involves the **proportional relationship between a part and the whole**.
  - Statement 2 involves the **distinction between a percentage (proportion) and a fixed number**, i.e., the pattern of difference between a statistical regularity (average fault rate) and a deterministic event ("exactly").
  - Statement 3 involves the fundamental probability pattern of **equivalence between long-term frequency and single-event probability**.

5th Place: Aspect 7 – Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- **Analysis**: To judge the statements, students need to simplify the real-world production context into clear mathematical questions.
- **Manifestation in the Question**:
  - Simplifying "One third of the players..." to calculating `2000/(2000+6000)`.
  - Simplifying "In each batch of 100..." to understanding that "5% is an average expectation, not a precise outcome every time."
  - Simplifying "probability that it will need repair" to directly equating it with the "average percentage faulty" given in the table, i.e., `3% = 0.03`.

6th Place: Aspect 12 – Identify the constraints, assumptions simplifications in a mathematical model

- **Analysis**: The question implies important statistical assumptions. Students need to identify these assumptions to make correct judgments.
- **Manifestation in the Question**: The most crucial aspect is identifying the **meaning of "average percentage" and the assumptions for its use**. Judging Statement 2 particularly depends on this: one must recognize the assumption that "5% is a long-term average," which does not imply deterministic outcomes. Statement 3 also assumes "random selection" and a stable fault rate.

7th Place: Aspect 2 – Identify the key variables in a model

- **Analysis**: This is closely related to Aspect 6 but focuses more on the identification of the variables themselves. When constructing the model, students must first determine which numerical values (variables) are important.
- **Manifestation in the Question**: As described under Aspect 6, the four core data points (2000, 6000, 5%, 3%) are the key variables for solving this problem.

8th Place: Aspect 9 – Translate a problem into a standard mathematical representation or algorithm

- **Analysis**: For some statements, the judgment process can be translated into standard calculations.
- **Manifestation in the Question**:
  - Statement 1 is translated into a fraction calculation and comparison.
  - Statement 2 is translated into an understanding of the concept of "percentage," requiring no complex algorithm.
  - Statement 3 is a direct conceptual correspondence.

第1位：方面 4 - 阅读、解码并理解表述、问题、任务、对象或图像，以建立情境模型

- **分析**：这是学生接触该题的第一步也是最基础的一步。他们必须阅读表格数据（日均产量、平均故障百分比）和三个陈述句，并理解每个陈述在具体情境（Electrix公司生产）中的含义。
- **题目体现**：学生需要解码“one third”（三分之一）、“exactly 5 out of 100”（每100个中恰好5个）、“probability...is 0.03”（概率为0.03）等表述，并将其与表格中的具体数字（2000, 6000, 5%, 3%）联系起来，从而在心中构建一个关于产量分布与故障率的数学模型。

第2位：方面 1 - 选择描述问题的数学描述或表征

- **分析**：在理解情境后，学生需要为每个陈述选择合适的数学工具或概念进行判断。
- **题目体现**：
  - 对于陈述1（视频播放器是否占日产量1/3），需要选择“比例/分数”这一数学描述：视频播放器占比 = 2000 / (2000+6000) = 1/4，而非1/3。
  - 对于陈述2（每100个视频播放器中是否恰好5个故障），需要选择“百分比的意义”这一描述：5%是长期统计的平均值，并不意味着每个100的批次中恰好有5个。
  - 对于陈述3（随机抽取一个音频播放器需要维修的概率是否为0.03），需要选择“概率的统计定义”这一描述：故障率3%可直接解释为单个随机产品的故障概率。

第3位：方面 6 - 识别并描述现实问题中的数学方面，包括识别重要变量

- **分析**：学生需要从生产情境中抽象出关键的数学变量，并理解它们之间的关系。
- **题目体现**：关键变量包括：视频播放器日产量（`V=2000`）、音频播放器日产量（`A=6000`）、视频故障率（`p_v=5%`）、音频故障率（`p_a=3%`）。陈述的判断正是基于对这些变量及其含义的准确识别与运用。

第4位：方面 5 - 在问题或情境中识别数学结构（规律、关系、模式）

- **分析**：题目要求学生看出数据背后隐含的数学关系和模式。
- **题目体现**：
  - 陈述1涉及 **部分与整体的比例关系。**
  - 陈述2涉及 **百分比（比例）与确定数量之间的区别**，即统计规律（平均故障率）与确定性事件（“exactly”）之间的差异模式。
  - 陈述3涉及 **长期频率与单次事件概率的等价关系** 这一概率论基本模式。

第5位：方面 7 - 简化或分解情境，使其易于进行数学分析

- **分析**：为了判断陈述，学生需要将现实生产情境简化为清晰的数学问题。
- **题目体现**：
  - 将“One third of the players...”简化为计算 `2000/(2000+6000)`。
  - 将“In each batch of 100...”简化为理解“5%是平均期望，不是每次精确结果”。
  - 将“probability that it will need repair”简化为直接等同于表格中给出的“average percentage faulty”，即 `3% = 0.03`。

第6位：方面 12 - 识别数学模型中的约束、假设与简化

- **分析**：题目隐含了重要的统计假设，学生需要识别这些假设才能做出正确判断。
- **题目体现**：最核心的是识别 **“平均百分比”的含义及其使用假设**。陈述2的判断尤其依赖于此：必须识别出“5%是长期平均值”这一假设，它并不意味着确定性的结果。同时，陈述3也假设了“随机选择”且故障率稳定。

第7位：方面 2 - 识别模型中的关键变量

- **分析**：这与方面6紧密相关，但更侧重于变量本身的识别。学生在构建模型时，必须首先确定哪些是重要的数值（变量）。
- **题目体现**：正如在方面6中所述，四个核心数据（2000, 6000, 5%, 3%）就是解决此题的关键变量。

第8位：方面 9 - 将问题转化为标准的数学表征或算法

- **分析**：对于一些陈述，其判断过程可以转化为标准计算。
- **题目体现**：
  - 陈述1转化为分数计算并比较。
  - 陈述2转化为对“百分比”概念的理解，无需复杂算法。
  - 陈述3直接是概念对应。